

Enhancing Perimeter Protection using Φ -OTDR and CNN for Event Classification

Adrian Tomasov,^{1*} Pavel Zaviska,¹ Vladimir Spurny,¹ Petr Dejdar,¹ Petr Munster,¹
Tomas Horvath,¹ Ondrej Klicnik¹

¹Brno University of Technology, Faculty of Electrical Engineering and Communications, Dept. of
Telecommunications, Technická 12, 616 00 Brno, Czech Republic

*tomasov@vutbr.cz

Abstract: The paper introduces an enhanced method combining Φ -OTDR and CNNs for an accurate object classification for perimeter protection. The proposed approach achieves an accuracy of 91 % of common events in the optical fiber vicinity. © 2023 The Author(s)

1. Introduction

Perimeter protection is crucial for safeguarding critical infrastructure, facilities, and assets. Traditional perimeter security systems often rely on cameras, motion detectors, and fences, which may have limitations in certain environments or face challenges with false alarms. Distributed acoustic sensing (DAS) using phase-sensitive optical time-domain reflectometry (Φ -OTDR) in intensity mode offers an alternative approach that harnesses the power of fiber-optic technology to enhance perimeter protection systems.

DAS systems based on Rayleigh scattering have found diverse applications across various domains, including pipeline monitoring, railway monitoring, urban safety, earthquake monitoring, and perimeter security and protection [1, 2]. Fiber-optic sensors provide new avenues for advancements by integrating modern machine learning techniques [3]. Particularly, combining DAS with convolutional neural networks (CNNs) has shown promising results in several perimeter protection and intrusion detection applications. These applications include high-speed railway intrusion detection, classification of loud objects in fiber surroundings, and denoising of DAS data [4–6].

This paper aims to build upon the existing research and propose an enhanced method with a CNN optimized for event and object classification using Φ -OTDR data deployed as an optical sensor for a campus-wide intrusion detection system. We collect a comprehensive dataset of Φ -OTDR captured in various perimeter protection scenarios, including different event types and object categories. The dataset will be used to train and evaluate our proposed CNN model. The ultimate goal is to advance the field of perimeter protection by developing a reliable and efficient system capable of accurately detecting and classifying events and objects.

2. Methodology

The proposed methodology for campus perimeter protection and object detection involves deploying a Φ -OTDR system for distributed acoustic sensing in fiber surroundings. This system utilizes advanced fiber-optic technology to capture acoustic data along the protected perimeter. The collected data is then analyzed using a CNN to identify and classify abnormal objects detected at each sensing point.

By leveraging the capabilities of Φ -OTDR, which provides detailed information about fiber characteristics and detects anomalies, the system can effectively monitor the perimeter and identify potential threats. The CNN plays a crucial role in analyzing the captured data and accurately distinguishing between normal and abnormal objects based on their unique acoustic signatures. This combined approach of Φ -OTDR and CNN ensures a robust and efficient perimeter protection system that can detect and classify various types of objects, enhancing the overall security of the campus.

2.1. Distributed acoustic sensing – Φ -OTDR

Φ -OTDR sends laser pulses into an optical fiber and analyzes the backscattered light. The pulses travel within the fiber, interacting with various phenomena such as reflections or scatterings. By measuring the time delay and phase shift of the Rayleigh backscattered light, the Φ -OTDR system can determine the characteristics of the fiber and identify any anomalies.

Reflectometry methods provide essential information about the fiber characteristics, including the length, attenuation, and identification of events such as fiber breaks, bends, or other impairments. By analyzing the phase shift and time delay of the backscattered light, Φ -OTDR enables accurate fault localization and precise measurement of fiber properties. It plays a crucial role in the maintenance and troubleshooting of optical networks, ensuring their reliable performance.

2.2. Dataset acquisition and preprocessing

The dataset was collected from a dedicated experimental fiber-optics path for perimeter protection system. Situated half a meter beneath a publicly accessible road near the university campus (depicted in Figure 1), the fiber path benefits from high usage by students and visitors, resulting in a substantial amount of data. The dataset encompasses a range of events and objects, including cars, longboards, instances of fiber manipulation, opening and closing of the manhole cover (housing fiber splices), running individuals, walking individuals, and a regular situation with no events. Table 1 provides the sample quantities for each labeled category within the dataset. This dataset comprehensively represents potential scenarios encountered in perimeter protection applications. It is a valuable resource for training and evaluating event and object classification algorithms that leverage Φ -OTDR technology and CNN.

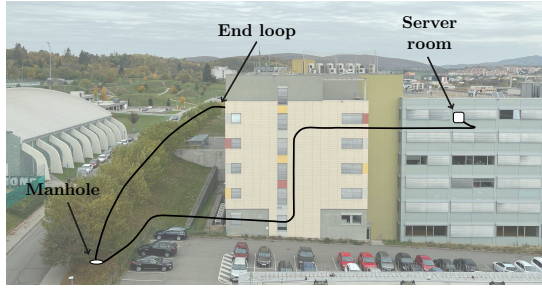


Fig. 1: The protected perimeter secured by a telecommunication optical fiber analyzed by Φ -OTDR.

Table 1: The number of samples per subset and label.

Label / Subset	Train.	Valid.	Test.	Total
Car	905	110	109	1124
Longboard	4578	568	570	5716
Fiber manipulation	9561	1198	1194	11953
Open/close manhole	2117	256	261	2634
Regular	3362	420	419	4201
Running person	5044	631	627	6302
Walking person	1952	241	242	2435
Total	27519	3424	3422	34365

The fiber path was analyzed using an Optasense ODH-F interrogator in intensity mode. The settings included a pulse rate of 20000 pulses per second, a system resolution of 1 meter per channel, and a maximum of 1700 channels. The raw output signal was unsuitable for direct analysis. Therefore, the magnitude spectrum using an oversampled discrete Fourier transform (DFT) with an oversampling factor of 3 for enhanced frequency resolution was computed for each time-window of 8192. This resulted in a three-dimensional magnitude spectrum: $channel \times time-window \times frequency$. To reduce redundancy and optimize the representation, only first 2048 elements of the single-sided magnitude spectrum was selected, eliminating the higher frequencies since the lower frequency range contained the most relevant information for analysis [7].

To facilitate manual labeling, the data was converted into a human-readable format. This involved calculating the estimate of the bandlimited signal power in the frequency range of 20 to 400 Hz using power spectral density (PSD). The resulting two-dimensional matrix (as depicted in Figure 2a) represented the labeled data, which was manually annotated using a graphical tablet. The annotations provided indices corresponding to event occurrences within the original dataset, as shown in Figure 2b. Certain labeled vectors exhibited significant levels of environmental noise and were subsequently excluded from the dataset.

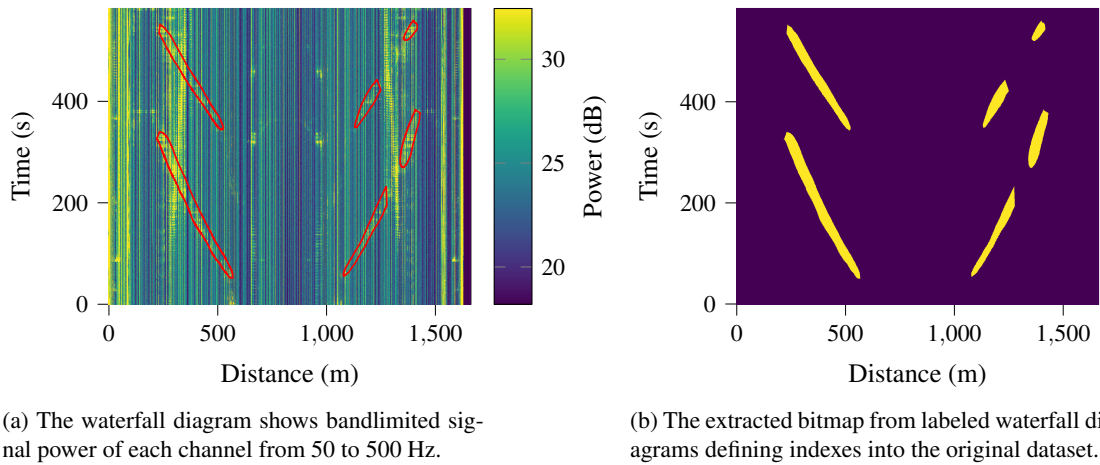


Fig. 2: The captured datasets are analyzed by evaluating the average power spectral density of a certain frequency range used for manual dataset labeling.

2.3. Neural network architecture

The proposed neural network consists of two convolutional layers [8] with 64 and 256 filters, respectively. Each convolutional layer is followed by a leaky rectifier linear unit activation function (LeakyReLU) [9] and a max pooling layer with a pool size equal to 4. The output of convolutional layers is flattened into a vector processed by a dense layer with 1024 with a *sigmoid* activation function and an output dense layer with seven neurons and the *softmax* function due to the classification problem. A schematic diagram of the proposed neural network with data shapes is shown in Figure 3, where purple rectangles stand for layers, red for activation functions, green for vector modification, and yellow for input and output.

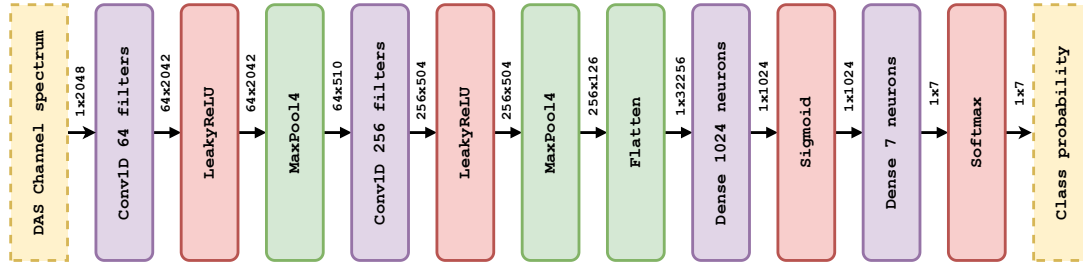


Fig. 3: The schematic diagram of the proposed convolutional neural network for event classification.

3. Learning and results

The training took 100 epochs, derived from several experiments with an early stopping mechanism engaged. The grid-search algorithm estimated the optimal training parameters, such as the batch size of 256 elements, learning rate 0.001, and reducing learning by 0.1 after 50 epochs [10]. The training used *Adam* optimizer [11] and *cross-entropy* loss function [12] with class weights to compensate the dataset imbalance problem. 10-fold cross-validation omits the dataset selection error [13]. The evolution of loss and accuracy over epochs on training and validation datasets is shown in Figure 4, where the solid lines represent a mean of values for the epoch, and the background fill represents a standard deviation of the metric.

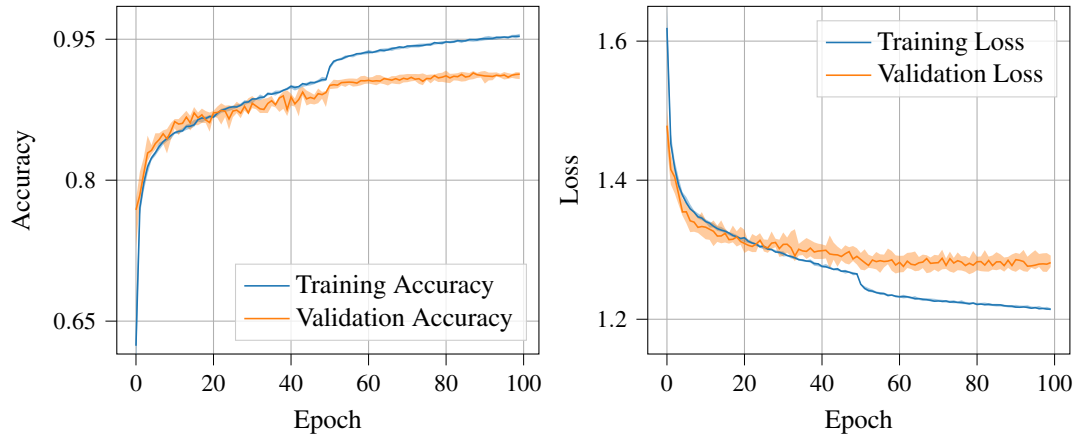


Fig. 4: Accuracy and loss metrics over training epochs evaluated on training and validation dataset.

The final model has an accuracy of about 95 % on the training dataset and about 91 % on the validation dataset. The evaluation of the testing dataset is shown in Figure 5, where the accuracy is exactly 91.41 %. Most of the classification classes have very high detection precision, except the car class. Many vectors from the car class were filtered out during preprocessing, and the remaining samples have a low signal-to-noise ratio. On the other hand, the model has an outstanding performance in deciding between *regular* and *irregular* (different than regular) because it misclassified only a single sample on the testing dataset.

4. Conclusion

Traditional security systems have limitations, leading to the exploration of alternative approaches such as DAS using Φ -OTDR in intensity mode. This paper aims to advance perimeter protection by proposing an enhanced method that combines DAS with CNNs for event and object classification. The study utilizes a comprehensive

True label	Car	77	1	0	4	0	12	15
	Longboard	4	541	1	7	0	17	0
	Manip.	11	5	1096	29	0	26	27
	Manhole	6	5	4	204	0	23	19
	Regular	0	0	0	0	418	0	1
	Running	16	3	7	11	0	572	18
	Walk	5	0	3	3	0	11	220
		Predicted label						
		Car	Longboard	Manip.	Manhole	Regular	Running	Walk

Fig. 5: Confusion matrix generated using testing subset classification results.

dataset of Φ -OTDR captured in various perimeter protection scenarios. The proposed CNN model, trained on this dataset, shows promising results in accurately detecting and classifying events and objects. The integration of fiber-optic technology with modern machine learning techniques holds great potential for enhancing perimeter protection systems.

Acknowledgment

Research described in this paper was supported by the Ministry of the Interior of the Czech Republic, program OPSEC, under grant VK01030121, project *Middle range distributed fiber-optic sensing system for acoustic vibration and temperature monitoring on critical infrastructures*.

References

1. Y. Shang, M. Sun, C. Wang, J. Yang, Y. Du, J. Yi, W. Zhao, Y. Wang, Y. Zhao, and J. Ni, "Research progress in distributed acoustic sensing techniques," *Sensors* **22** (2022).
2. Z. He, Q. Liu, X. Fan, D. Chen, S. Wang, and G. Yang, "Fiber-optic distributed acoustic sensors (DAS) and applications in railway perimeter security," in *Advanced Sensor Systems and Applications VIII*, vol. 10821 T. Liu and S. Jiang, eds., International Society for Optics and Photonics (SPIE, 2018), p. 1082102.
3. M. Furdek, C. Natalino, F. Lipp, D. Hock, A. D. Giglio, and M. Schiano, "Machine learning for optical network security monitoring: A practical perspective," *J. Light. Technol.* **38**, 2860–2871 (2020).
4. Z. Li, J. Zhang, M. Wang, J. Chai, Y. Wu, and F. Peng, "An anti-noise ϕ -OTDR based distributed acoustic sensing system for high-speed railway intrusion detection," *Laser Phys.* **30**, 085103 (2020).
5. Y. Zhao, Y. Li, and N. Wu, "Distributed acoustic sensing vertical seismic profile data denoiser based on convolutional neural network," *IEEE Trans. on Geosci. Remote. Sens.* **60**, 1–11 (2022).
6. Z. Li, J. Zhang, M. Wang, Y. Zhong, and F. Peng, "Fiber distributed acoustic sensing using convolutional long short-term memory network: a field test on high-speed railway intrusion detection," *Opt. Express* **28**, 2925–2938 (2020).
7. P. Dejdar, P. Závíška, S. Valach, P. Münster, and T. Horváth, "Image edge detection methods in perimeter security systems using distributed fiber optical sensing," *Sensors* **22**, 4573 (2022).
8. A. Ajit, K. Acharya, and A. Samanta, "A review of convolutional neural networks," in *2020 international conference on emerging trends in information technology and engineering (ic-ETITE)*, (IEEE, 2020), pp. 1–5.
9. S. Mastromichalakis, "ALReLU: A different approach on Leaky ReLU activation function to improve neural networks performance," *arXiv preprint arXiv:2012.07564* (2020).
10. R. Ge, S. M. Kakade, R. Kidambi, and P. Netrapalli, "The step decay schedule: A near optimal, geometrically decaying learning rate procedure for least squares," *Adv. neural information processing systems* **32** (2019).
11. D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *3rd International Conference on Learning Representations, ICLR 2015*, (2015).
12. A. Semenov, V. Boginski, and E. L. Pasiliao, "Neural networks with multidimensional cross-entropy loss functions," in *International Conference on Computational Data and Social Networks*, (Springer, 2019), pp. 57–62.
13. T.-T. Wong, "Performance evaluation of classification algorithms by k-fold and leave-one-out cross validation," *Pattern Recognit.* **48**, 2839–2846 (2015).